Market study plan: a probiotic tomato-based drink

Hatef Tabbakhian · November 2026 · public data only · code and data: github.com/Leotaby/tomato-probiotic-drink-market-study

A probiotic tomato drink can start where tomato juice is already a habit: **Germany buys 70% of Italy's tomato-juice exports**. It can also try to build a new wellness category in Italy. The study tests both routes and prices the probiotic claim with a choice experiment.

QUESTIONS THE STUDY ANSWERS

1. **Where** is the demand: Italy, or export markets with a tomato-juice habit?
2. **Who** buys: which segments, how many?
3. **What** sells it: probiotic claim, origin, organic, salt, and the premium people accept.
4. **How much:** realistic volumes and revenues, under explicit scenarios.

WHAT PUBLIC DATA ALREADY SHOWS

- Italy's tomato-juice exports: **EUR 4.8 m (2010) → EUR 14.2 m (2025)**, volume nearly flat (12.4 → 13.3 thousand t). Value per kg EUR 0.39 → 1.06: partly up-market, partly raw-material costs (tomato prices +50% in four years, ANICAV).
- **Germany 70.5%** of Italy's exports, South Korea 11.3%, France 8.7%; only 16.5% leaves the EU.
- Italy is the **EU's second exporter** (24%, after Spain); Germany is the largest importer; Italy imports almost none.
- Italy processed **5.8 m t** of tomatoes in 2025, second in the world (ANICAV).
- Plant-based probiotic juices kept **> 6 log CFU/mL for 33 days** refrigerated, liked 6.9–8.9/9 (Condurso et al., 2025): chilled distribution.
- Label: Italy allows "probiotico" if a daily serving has at least 10^9 live cells (6.6 log CFU/mL in a 250 ml bottle); no EU health claim is authorised, so test only usable wording.

SEVEN-MONTH PLAN

Month	Work → output
1	Desk research, price checks in Italy and Germany, top-down sizing → price map
2	Survey and choice-experiment design with the lab and producers; pilot (n ≈ 30)
3–4	Online panel: Italy n = 500, Germany n = 300; tasting session with the prototype (n = 60–80)
5	Choice models, purchase intention, segments, sensory analysis → WTP per attribute
6	Bottom-up market model with Monte Carlo ranges, by country and channel
7	Report and presentation to the producers; draft paper

CHOICE EXPERIMENT

Attribute	Levels
Price, 250 ml	EUR 1.50 · 2.00 · 2.50 · 3.00
Probiotic	none · "with live probiotic cultures"
Origin	not stated · Italian · Campanian tomato
Organic · Salt	no / yes · standard / no added salt

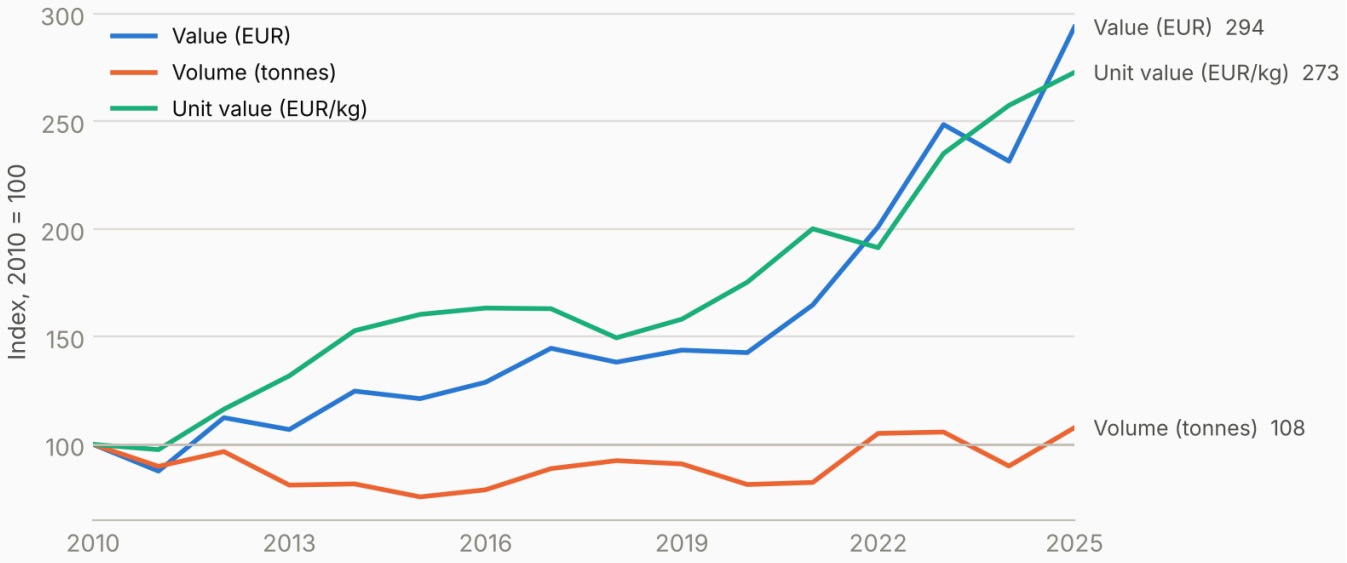
8 tasks × (2 products + "neither"), 3 blocks. **Models:** multinomial logit → mixed logit and latent class; WTP = $-\beta_{attr}/\beta_{price}$. Also ordered logit (purchase intention), quantile regression (stated WTP), ANOVA and penalty analysis (tasting).

Sample size: simulation from assumed preferences recovers the probiotic premium within ±EUR 0.14 (150 respondents), ±0.09 (300), ±0.08 (500) → **at least 300 per country**.

Sources: Eurostat Comext DS-045409, product 200950 (extracted 8 Nov 2026); ANICAV, 1 Dec 2025; Condurso et al., Foods 14(12):2148, 2025; Ministero della Salute, guidelines on probiotics (rev. March 2018); Reg. (EC) 1924/2006. Trade code 200950 is a proxy (unfermented juice) and trade values are not retail sales.

Italy's tomato-juice exports: value up almost 3x since 2010, volume flat

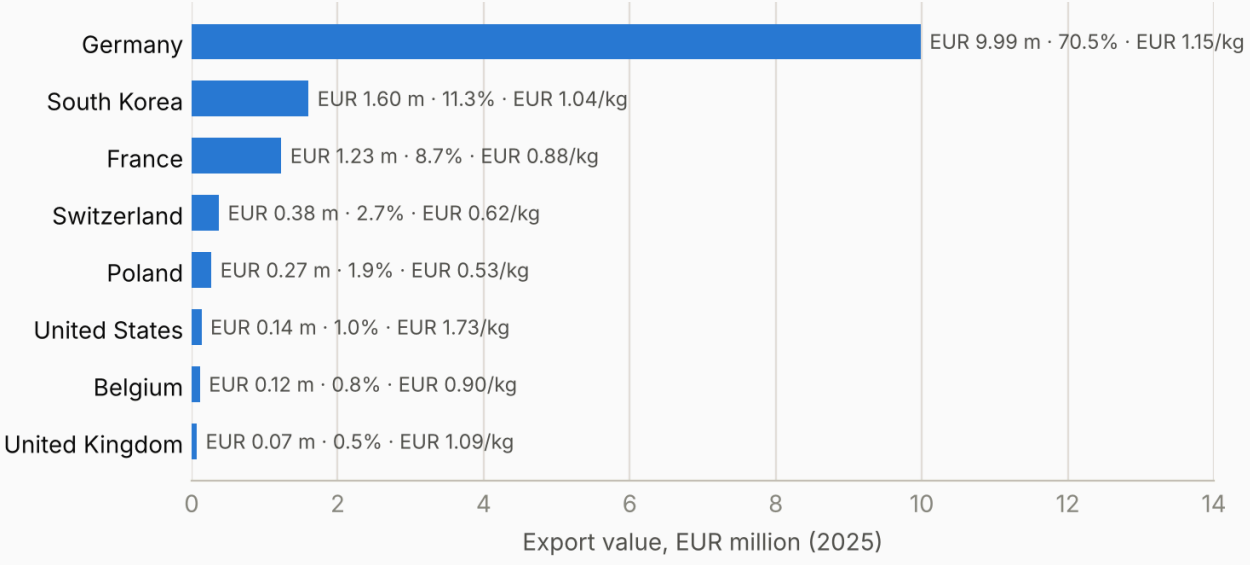
Exports EUR 4.8 m → EUR 14.2 m; unit value EUR 0.39 → 1.06 per kg



Source: Eurostat Comext DS-045409, product 200950 (tomato juice, unfermented); extracted 8 Nov 2026.

Seven in ten euros of Italy's tomato-juice exports go to Germany

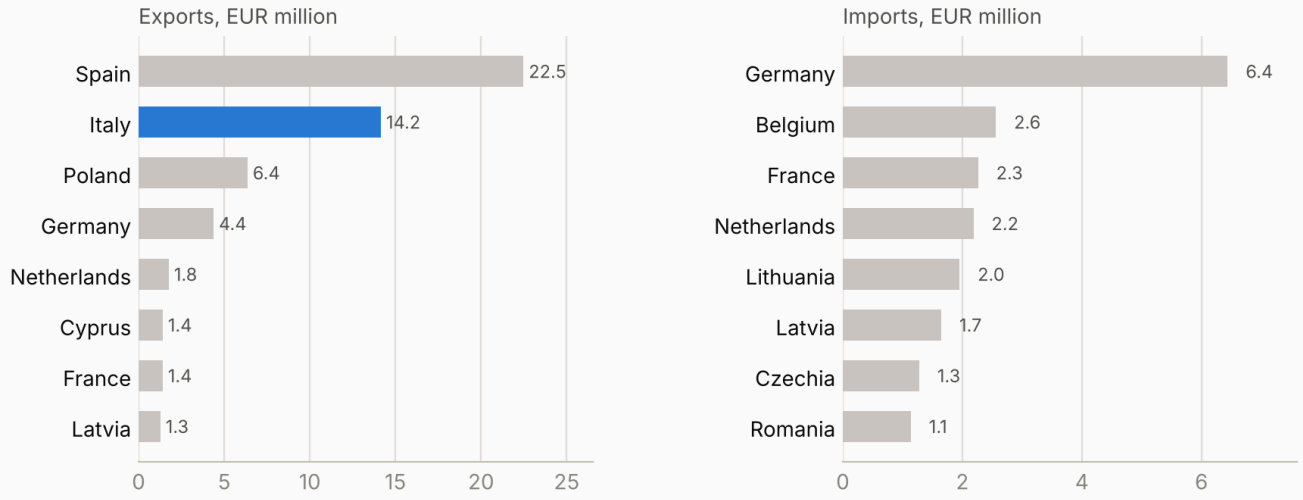
Top destinations, 2025 (total EUR 14.2 m). Labels: value · share · unit value



Source: Eurostat Comext DS-045409, product 200950 (tomato juice, unfermented); extracted 8 Nov 2026.

Spain and Italy supply Europe's tomato juice; Germany is the largest buyer

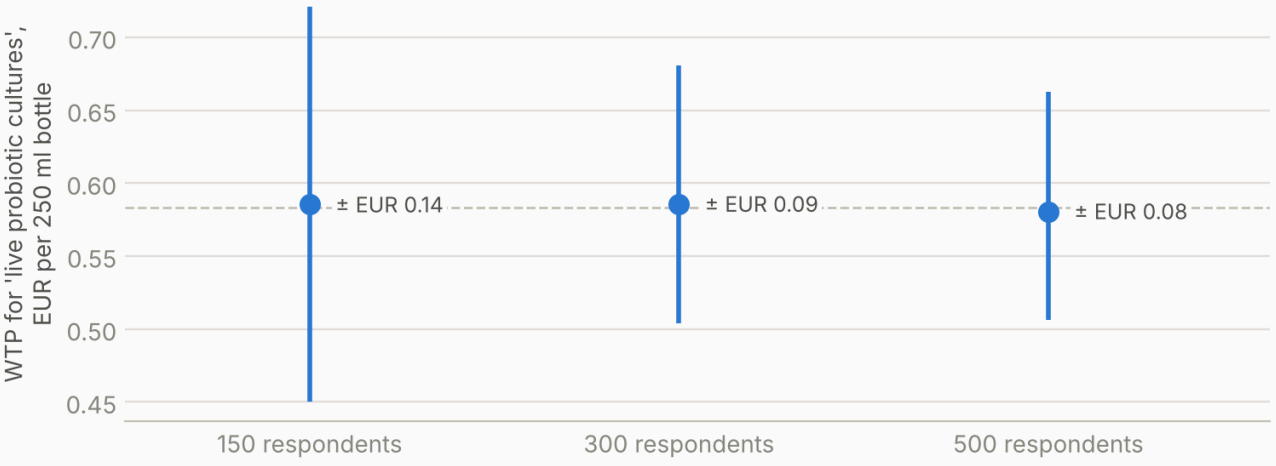
EU member states' trade in tomato juice with all partners, 2025 (top 8 each)



Source: Eurostat Comext DS-045409, product 200950 (tomato juice, unfermented); extracted 8 Nov 2026.
Mirror flows differ: Germany reports EUR 6.4 m imports from all partners, while Italy reports EUR 10.0 m exports to Germany.

Design check: how precisely the survey would measure the probiotic premium

Simulated choices from assumed preferences · 300 replications per sample size · 8 tasks per respondent · multinomial logit



Bars: 95% range of the WTP estimates across replications. Dashed line: assumed true WTP (EUR 0.58).
Planning exercise on simulated data, not a result about consumers.